



Cambridge IGCSE™

DESIGN AND TECHNOLOGY

0445/32

Paper 3 Resistant Materials

May/June 2026

MARK SCHEME

Maximum Mark: 50

Published

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

This document consists of **11** printed pages.

General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: - the specific content defined in the mark scheme - the specific skills defined in the mark scheme or in the level descriptions - the standard of response required as exemplified by the standardisation scripts.
2	Examiners must award whole marks (not half marks, or other fractions).
3	Examiners must award marks positively . This means that: - marks are not deducted for errors or omissions - marks are awarded for correct/valid answers as defined in the mark scheme and also for other valid answers which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate - the quality of spelling, punctuation and grammar are judged only when the mark scheme indicates that these features are specifically assessed in the question. However, the meaning of all responses must be unambiguous.
4	Examiners must consistently apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the full range of marks defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and must not award marks with grade thresholds or grade descriptions in mind.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	Unclear
	Benefit of the doubt
	Incorrect point
	Error carried forward
N/A	Highlighting areas of text
	No benefit of doubt given
N/A	Off-page comment – allows comments to be entered off the page
	Repeat
	Indicates that the point has been noted, but no credit has been given
	Indicates that the point has been noted, but no credit has been given (big)

Annotation	Meaning
	Correct point
	Too vague
	Relevant detail

Question	Answer	Marks	Guidance
1	Any 3 benefits such as: lightweight, smooth, colourful, easily moulded, easily [mass] produced, durable, self-finished, waterproof, does not corrode, easy to clean, hardwearing, does not splinter, tough [3 × 1]	3	Accept benefits to manufacturers Do not accept: cheap, low production costs, comfortable

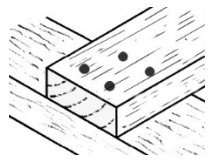
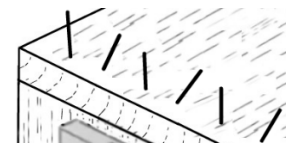
Question	Answer	Marks	Guidance
2	Odd number of fingers [1] accurate width of fingers [1] accurate depth of fingers [1]	3	

Question	Answer	Marks	Guidance
3	A round [1] B countersunk / countersink [1]	2	Do not accept any other variations

Question	Answer	Marks	Guidance
4(a)	odd-legs, scribe, engineer's square, try square, centre punch, dot punch	1	Accept scribe Do not accept: marker pen
4(b)	scribe, engineer's square, try square	1	

Question	Answer	Marks	Guidance
5	Any 2 properties: lightweight, moisture resistant, good heat retention, recyclable, good chemical resistance, hygienic, non-toxic, cheap, easily formed [2 × 1]	2	Accept properties for manufacturers Do not accept: easy to clean, flexible, durable

Question	Answer	Marks	Guidance
6	Thermochromic ink/pigment/plastic/paint [1]	1	Accept Thermochromic on its own

Question	Answer	Marks	Guidance
7(a)		2	Minimum 3 nails
7(b)		2	Award 1 mark for nails shown in 'sloping' positions

Question	Answer	Marks	Guidance
8	A tube / tubing [1] B strip [1] B bar or rod [1]	3	Accept hollow tube

Question	Answer	Marks	Guidance
9	2 hinges: butt, piano, flush, butterfly [2 × 1]	2	

Question	Answer	Marks	Guidance
10	Some sort of straight edge/blade [1] Named materials [1] Joined to holder [1]	3	

Question	Answer	Marks	Guidance
11(a)	Any 2 benefits such as: durable, attractive, finishes well, sustainable, tough [2 × 1]	2	Accept any valid benefits
11(b)(i)	Bevel [1]	1	
11(b)(ii)	Square [1]	1	
11(b)(iii)	Any 2 safety precautions such as: wear eye protection, no loose clothing, work piece flat on table, fingers behind sanded surface, dust extraction, wear an apron [2 × 1]	2	Accept any valid precautions
11(c)(i)	Minimum Ø 6 mm – maximum Ø 15 mm	1	
11(c)(ii)	Centre line drawn 15 mm from edge [1] Top and bottom holes no closer than 5mm from end of area [1]	2	
11(d)	Accurate ‘projection’ of tenons down to mortises [2 × 1] Lines drawn showing some ‘depth’ of mortises [1]	3	
11(e)	Practical method: some sort of ‘lapping’ [1] Named materials [1] Constructional details [1]	3	Named materials must be specific, NOT generic
11(f)(i)	Cramp shown in position between platform and base [1] ‘Angled’ scrap wood shown against platform [1] Scrap wood shown under base [1]	3	
11(f)(ii)	Sash cramp, ‘F’ cramp, Quick/Speed cramp, G cramp	1	Accept any valid methods
11(g)(i)	To avoid scratches, damage, splinters	1	Do not accept: splitting

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Question	Answer	Marks	Guidance
11(g)(ii)	Any 2 problems: brush strokes visible, excessive or uneven application of finish, drips, bristles left on surface of wood [2 × 1]	2	
11(h)	Takes up less space, can be dismantled and folded flat, painted finish to 'match' environment, holds book securely, cheaper, flat pack design, modern design, smaller/ more compact, holds pages open, simple design [3 × 1]	3	Do not accept: easier to clean, lighter

Question	Answer	Marks	Guidance
12(a)	Any 2 properties: attractive, easy to shape, inherent colour, self-finished, impact resistant, easy to cut and shape, different colours available, durable, easy to clean, smooth to touch [2 × 1]	2	Accept any valid properties relating to the clock Do not accept: lightweight, does not corrode
12(b)(i)	Marks made by a marker pen can be erased / Marks made by a scribe cannot / Marker pen more easily seen.	1	
12(b)(ii)	Dividers	1	

Question	Answer	Marks	Guidance
12(b)(iii)	Method 1 Drill hole in acrylic [1] Insert blade of coping, Hegner, scroll saws and cut out waste [1] Use of files / bobbin sander to make sawn edges smooth [1] Accuracy of named tools mark can be awarded if ALL 3 stages are given [1] Method 2 Hole saw named [1] Sketch of hole saw [1] Acrylic held securely [1] Sacrificial board [1]	4	Laser cutter award 1 mark only
12(c)	Use of line bender/strip heater/ hot air gun to soften acrylic [1] Bent to shape using a former/jig or setting the line bender machine mechanism [1] Method of retention while acrylic cools [1]	3	Do not accept: acrylic heated in an oven but candidates can access marks for 'bending' and 'retention'
12(d)	3 stages: 1 [draw] filing or use of 'scraper' [1] 2 use of wet and dry [silicon carbide] paper [1] 3 use of polishing mop and compound or polishing by hand [1]	3	Correct sequence NOT required Accept abrasive paper Do not accept: use of emery cloth
12(e)	Danger: inhalation of fumes. Precaution; wear a face/mouth mask [2 × 1] Danger: skin irritant. Precaution: wear gloves or use barrier cream [2 × 1] Danger: fire hazard. Precaution: good ventilation [2 × 1]	4	3 possible dangers. Only 2 required

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Question	Answer	Marks	Guidance
12(f)	Sketch to show use of cramps to hold parts together [1] Use of scrap wood to distribute pressure/prevent damage to surfaces of acrylic [1]	2	
12(g)	Some sort of practical 'stand' [0–2] Stand joined to clock using acrylic adhesive [1] Details of constructions [0–2]	5	

Question	Answer	Marks	Guidance
13(a)(i)	Wide variety of softwoods available. Red deal, redwood, pine, whitewood, cedar, parana pine	1	Check names of obscure trees
13(a)(ii)	Plywood, blockboard, laminboard, MDF, furniture board	1	Do not accept chipboard, plyboard
13(a)(iii)	Softwood is generally cheaper, more readily available	1	Do not accept easier to work
13(b)	Any 2 reasons for popularity: value for money, cheaper than ready-assembled products, satisfaction with self-assembly process, some products can be taken apart for storage, taken home easily, cheaper [implied comparison] [2 × 1]	2	Accept any valid reasons Do not accept easy to construct
13(c)(i)	A tap wrench [1] B tap, taper tap [1]	2	
13(c)(ii)	Workpiece held securely Tap tightened /located in tap wrench Tap inserted into drilled hole perpendicular Pressure exerted while turning tap clockwise and return tap 1/2 anti clockwise to break off cut metal [4 × 1]	4	

Question	Answer	Marks	Guidance
13(d)	140mm length shown without measuring: use of rule, index mark, 'stop' [1] Softwood held securely while strips are cut to length [1] Description of use [0–2]	4	Do not accept use of circular or band saws Description of use to include e.g. tools and equipment used, materials, constructions
13(e)	Varnish, white polish, French polish, Danish oil, sealer, lacquer, shellac, linseed oil, wax [2 × 1]	2	Do not accept 'oil'
13(f)	2 precautions: good ventilation, wear a face mask/goggles, mouth mask, wear gloves, glasses [2 × 1]	2	
13(g)(i)	Cut out: use of tin snips, hacksaw [1] File to make sawn edges smooth and safe [1] Bend to shape: use of former/vice [1] Method of force: hammer, mallet [1]	4	Answers that use line bending as a method of bending sheet metal can achieve marks for use of a former and the method of force, hammer or mallet
13(g)(ii)	Wet and dry [silicon carbide] paper, emery cloth, aluminium oxide [2 × 1]	2	Do not accept steel wool